

CASE RECORDS of the MASSACHUSETTS GENERAL HOSPITAL

Founded by Richard C. Cabot
 Eric S. Rosenberg, M.D., *Editor*
 David M. Dudzinski, M.D., Meridale V. Baggett, M.D., Kathy M. Tran, M.D.,
 Dennis C. Sgroi, M.D., Jo-Anne O. Shepard, M.D., *Associate Editors*
 Emily K. McDonald, Tara Corpuz, *Production Editors*



Case 19-2023: An 80-Year-Old Man with Left Foot Drop

Paloma González-Pérez, M.D., Ph.D., Karen A. Buch, M.D.,
and Reza Sadjadi, M.D.

PRESENTATION OF CASE

Dr. Joshua U. Okonkwo (Neurology): An 80-year-old man was evaluated in the neurology clinic of this hospital because of left foot drop.

The patient had been in his usual state of health until 5 years before the current presentation, when he first noticed that he needed to lift up his left leg while walking in order to prevent the toes of his left foot from dragging on the ground. During the next 2 years, his ability to dorsiflex the left ankle decreased progressively. Three years before this presentation, he began to use an ankle-foot orthosis.

One year before this presentation, the patient was evaluated in a neurology clinic affiliated with another hospital. He reported left foot drop, as well as occasional falls that he attributed to poor balance. Examination reportedly revealed wasting in the muscles below the left knee, weakness with attempts at dorsiflexion and eversion of the left ankle, loss of pinprick sensation to the midshins in both legs, and decreased vibratory sensation in the toes of both feet. The blood levels of cobalamin, folate, thyrotropin, and glycated hemoglobin and the results of serum protein electrophoresis were reportedly normal. Tests for syphilis and antinuclear antibodies were negative, as were blood tests for antibodies to sulfate-3-glucuronidyl paragloboside and myelin-associated glycoprotein and a serum paraneoplastic autoantibody panel. The blood creatine kinase level was 456 IU per liter (reference range, 30 to 194) and the aldolase level 5.8 U per liter (reference value, <8.1). Imaging studies were obtained.

Magnetic resonance imaging (MRI) of the head was reportedly normal. MRI of the lumbar spine reportedly revealed diffuse disk bulging with facet hypertrophy at multiple levels, which was contributing to mild spinal stenosis. Needle electromyography (EMG) of the left leg reportedly showed changes consistent with moderate-to-severe, subacute-to-chronic radiculopathy at L5 and S1 on the left side.

The patient declined further workup at that time. One year later, his symptoms had not resolved, and he presented to the neurology clinic of this hospital for further evaluation.

On the current evaluation, the patient reported stable left foot drop and mild low-back pain without radiating symptoms. He had no bowel or bladder changes,

From the Departments of Neurology (P.G.-P., R.S.) and Radiology (K.A.B.), Massachusetts General Hospital, and the Departments of Neurology (P.G.-P., R.S.) and Radiology (K.A.B.), Harvard Medical School — both in Boston.

N Engl J Med 2023;388:2379-87.

DOI: 10.1056/NEJMcpc2211512

Copyright © 2023 Massachusetts Medical Society.

CME
at NEJM.org

no double vision or hearing changes, and no difficulty with breathing, chewing, or swallowing. His medical history included hypertension, dyslipidemia, and adenocarcinoma of the prostate, which had been treated with local radiation therapy 4 years before this presentation. Medications included aspirin, hydrochlorothiazide, lisinopril, metoprolol, and simvastatin. The patient was married, resided in New England, and was retired from work in the insurance industry. He had smoked one pack of cigarettes per day for 30 years but had quit smoking 30 years before this presentation. He drank four alcoholic drinks per day. His sister had breast cancer, and his father had died in his 70s from esophageal cancer.

On examination, the blood pressure was 140/60 mm Hg, the pulse 72 beats per minute, the respiratory rate 14 breaths per minute, and the oxygen saturation 100% while the patient was breathing ambient air. The body-mass index (the weight in kilograms divided by the square of the height in meters) was 26.1. He had a step-page gait pattern with prominent left foot drop. He could not walk on his left heel, but he was able to walk on his left toes and to walk with a tandem gait.

There was mild facial weakness, which was present on both sides but was more prominent on the right side, with an asymmetric smile and weak eye closure and cheek puffing. The optic disks had sharp margins. The pupils were round and reactive to light. Extraocular movements were full. There was no weakness or atrophy of the jaw, palate, or tongue. Hearing was grossly intact. Shoulder shrug was asymmetric, with less rise on the right side. A motor examination revealed wasting of the shoulder girdle, which was more prominent on the right side, and wasting of the calves. Tone was mildly decreased throughout, and there were no fasciculations. There was severe scapular winging, which was present on both sides but was more prominent on the right side. There was no action or percussion myotonia or paramyotonia.

Strength was assessed according to the Medical Research Council scale, which ranges from 0 (no visible muscle contraction) to 5 (normal strength). On the right and left sides, strength was 4 and 4+, respectively, on shoulder abduction; 5 and 3 on ankle dorsiflexion; 4+ and 4 on big-toe extension; 5 and 4+ on ankle inversion; and 5 and 4– on ankle eversion. On both sides,

strength was 4+ on hip flexion and was normal on neck flexion and extension, elbow flexion and extension, wrist flexion and extension, finger extension and abduction, thumb abduction, hip abduction, knee flexion and extension, and plantar flexion. Deep-tendon reflexes were 1+ at the biceps, 1+ at the triceps, trace at the brachioradialis, 2+ at the knees, and not elicited at the ankles. Plantar responses were flexor. There was loss of pinprick and temperature sensations to the midshins in both legs, as well as decreased vibratory sensation to the ankles. Imaging studies were obtained.

Dr. Karen A. Buch: MRI of the lumbar spine (Fig. 1) revealed a large, leftward, eccentric disk herniation at L5 and S1 that impinged on the exiting L5 nerve roots, with greater impingement on the left side than on the right side. There was narrowing of the left subarticular zone and left lateral recess that encroached on the descending left S1 nerve root. Facet hypertrophy and ligamentous infolding at multiple levels were contributing to mild central spinal stenosis.

Dr. Okonkwo: Additional testing was performed, and a diagnosis was made.

DIFFERENTIAL DIAGNOSIS

Dr. Paloma González-Pérez: This 80-year-old man presented with a 5-year history of progressive and painless left foot drop. Examination revealed weakness that involved not only distal left-leg muscles but also facial, periscapular, hip-flexor, and distal right-leg muscles. In constructing a differential diagnosis, I will begin by localizing the patient's symptoms and examination findings within the nervous system.

Do the patient's deficits localize to the central nervous system (CNS) or the peripheral nervous system (PNS)? CNS disorders affect the brain or spinal cord, whereas PNS disorders involve the anterior horns, nerve roots, plexus, peripheral nerves, neuromuscular junction, or muscles.

CNS DISORDERS

Although the most common causes of foot drop are PNS disorders, CNS processes that disrupt the upper-motor-neuron tract (anywhere from the medial homunculus of the primary motor cortex at the frontal lobe, where the leg is represented, to the ventral gray matter of the spinal

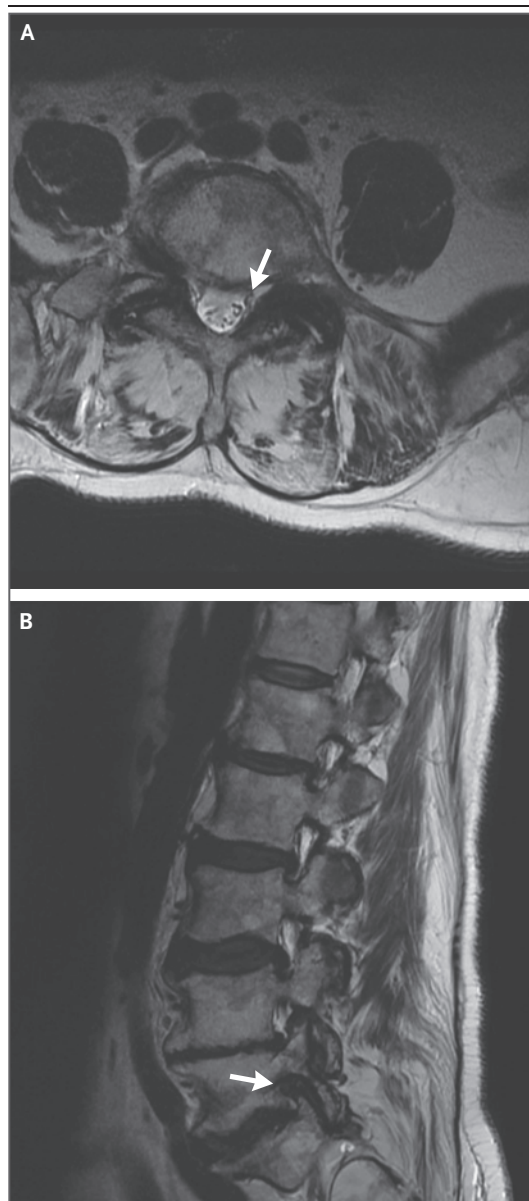


Figure 1. MRI of the Lumbar Spine.

A parasagittal T2-weighted image (Panel A) shows severe neural foraminal stenosis on the left side that is impinging on the exiting left L5 nerve root (arrow). An axial T2-weighted image (Panel B) shows a diffuse disk bulge that is in contact with, and is possibly impinging on, the descending left S1 nerve root (arrow); there is no high-grade central spinal stenosis at this level.

cord) should be considered.¹⁻⁶ However, such processes rarely cause foot drop in isolation. This patient did not have any upper-motor-neuron signs (such as increased muscle tone, clonus or brisk deep-tendon reflexes, or an extensor plantar response) that would suggest an insult to the

corticospinal tract in either the brain or the spinal cord. Moreover, a brain MRI was reportedly normal, and he did not have neck pain, sphincter dysfunction, or sensory loss at a level that would suggest myelopathy.

A CNS disorder that can mimic foot drop is dystonia, which can be the first manifestation of a parkinsonian syndrome.^{7,8} Dystonia is caused by dysfunction of the network involving the basal ganglia, cerebellum, and somatosensory cortex. In patients with dystonia that mimics foot drop, the results of brain imaging can be normal. However, muscle strength, muscle bulk, and deep-tendon reflexes would also be normal, and this patient had muscle weakness, muscle wasting, and hyporeflexia.

FOCAL PNS DISORDERS

Among causes of foot drop that localize to the PNS, focal disorders are more prevalent than generalized disorders. Focal PNS disorders include deep peroneal neuropathy, common peroneal neuropathy, sciatic neuropathy, lumbosacral plexopathy, and L5 radiculopathy. All these disorders can present with foot drop due to weakness of the tibialis anterior muscle, which is involved in ankle dorsiflexion.

Isolated neuropathy of the deep branch of the common peroneal nerve is unlikely in this patient because he had weakness on ankle eversion, an action performed by the peroneus longus and peroneus brevis muscles, which are innervated by the superficial branch of the common peroneal nerve. Common peroneal neuropathy is also unlikely because he had weakness on ankle inversion, an action performed by the tibialis posterior muscle, which is innervated by the tibial nerve.

Sciatic neuropathy with preferential involvement of the peroneal division that runs lateral to the tibial division (and is therefore more prone to injury) is a plausible explanation. However, this patient's 5-year history of progressive foot drop without reported sensory symptoms or pain would be atypical of sciatic neuropathy. In addition, sciatic neuropathy is often a complication of either trauma or hip surgery, neither of which had occurred in this patient.

The absence of pain and sensory symptoms would also be atypical of left lumbosacral plexopathy. An important exception is radiation-induced lumbosacral plexopathy, which is usually painless. However, symptoms in this patient

developed 1 year before treatment with radiation. L5 radiculopathy is a common cause of foot drop, and MRI of the lumbar spine in this patient revealed impingement on the left L5 nerve root. However, the absence of pain and sensory symptoms would be uncommon for L5 radiculopathy.

No focal PNS disorder would fully account for this patient's left-leg weakness. In addition, he had what I would describe as a "foot drop plus syndrome." I suspect that a generalized PNS disorder could provide a unifying diagnosis for both the left foot drop and the other muscle weakness noted on physical examination.

GENERALIZED PNS DISORDERS

Neuromuscular-Junction Disorders

Generalized PNS disorders can be further localized to the neuromuscular junction, nerve structures, or muscles. This patient's presentation is not consistent with a neuromuscular-junction disorder, such as myasthenia gravis or the Lambert-Eaton myasthenic syndrome. He did not have ocular symptoms or fluctuating weakness, findings that would be suggestive of myasthenia gravis, and he did not have fatigue, a common feature of the Lambert-Eaton myasthenic syndrome. In addition, the distribution of muscle weakness seen in this patient would be atypical of a neuromuscular-junction disorder. The muscle groups that were most severely affected in this patient were the periscapular muscles and the muscles in the anterior compartment of the left leg; this scapuloperoneal distribution of muscle weakness is suggestive of either a neurogenic disorder or a myopathic disorder.

Neurogenic Disorders

Neurogenic causes of scapuloperoneal syndrome include Charcot-Marie-Tooth disease type 1C and rare forms of spinal muscular atrophy.⁹⁻¹² Charcot-Marie-Tooth disease type 1C is unlikely in the absence of additional clinical features such as pes cavus deformity, hammer toe, sensory symptoms, and generalized areflexia. Spinal muscular atrophy with a scapuloperoneal distribution of muscle weakness is unlikely in the absence of skeletal dysplasia, vocal cord palsy, and muscle agenesis. Although progressive foot drop can be the first symptom of amyotrophic lateral sclerosis (ALS), a scapuloperoneal distribution of muscle weakness with

intact strength of the distal arms argues against a diagnosis of ALS.

Distinguishing Neurogenic from Myopathic Disorders

A finding on physical examination that can be helpful in differentiating neurogenic causes of foot drop from myopathic causes is the bulk of the extensor digitorum brevis muscle (located in the dorsum of the foot). In patients with neurogenic foot drop, length-dependent involvement of muscles innervated by the same nerve would be expected. The extensor digitorum brevis muscle and the tibialis anterior muscle (located in the anterolateral aspect of the shin) are both innervated by the deep branch of the common peroneal nerve. Atrophy of the extensor digitorum brevis muscle that precedes weakness of the tibialis anterior muscle would be suggestive of a neurogenic disorder. Normal bulk of the extensor digitorum brevis muscle in the presence of weakness of the tibialis anterior muscle is highly suggestive of a myopathic disorder. We do not know whether this patient had normal bulk of the left extensor digitorum brevis muscle, so I will rely on other clinical features to determine whether he had a neurogenic or myopathic disorder.

Scapuloperoneal syndromes due to neurogenic disorders are usually characterized by symmetric or slightly asymmetric muscle weakness. The strikingly asymmetric muscle weakness in this patient is more typical of scapuloperoneal syndromes due to myopathic disorders.

In addition, the anatomical progression of muscle weakness in this patient helps in distinguishing between neurogenic and myopathic disorders. Patients with adult-onset neurogenic scapuloperoneal syndromes typically have weakness that begins in distal muscles and progresses to involve proximal muscles (centripetal spreading); foot drop may develop at a young age, and weakness may advance from distal to proximal muscles over a period of decades. By contrast, foot drop developed in this patient in the eighth decade of life. I suspect that he had weakness that began in the facial and proximal muscles when he was much younger, but the symptoms either were unnoticed or did not cause enough interference with activities of daily living for him to seek medical attention. The weakness then progressed from proximal to distal muscles

(centrifugal spreading) over a period of decades, leading him to seek medical attention for foot drop at 75 years of age. Centrifugal progression is more typical of myopathy.

The asymmetry and possible centrifugal progression of this patient's muscle weakness point toward a myopathic disorder. However, the EMG findings did not appear to show early recruitment of motor-unit action potentials (with short duration and low amplitude), which would be suggestive of a myopathic process. Instead, the EMG findings were reportedly consistent with a neurogenic process such as L5–S1 radiculopathy. It is important to note that the absence of a myopathic pattern on EMG does not rule out myopathy. It can be difficult to identify a myopathic pattern on EMG in the presence of a superimposed neurogenic process (such as radiculopathy), and some myopathies may have a pseudoneurogenic pattern.^{13–15} Although L5–S1 radiculopathy may be present in this patient, I suspect that a myopathy is the main contributor to his foot drop.

Myopathic Disorders

Although most acquired and hereditary myopathies are associated with a proximal and relatively symmetric distribution of muscle weakness, some typically affect distal muscles and can be asymmetric. In particular, a scapuloperoneal distribution can be seen with facioscapulohumeral muscular dystrophy (FSHD), Emery–Dreifuss muscular dystrophy, limb-girdle muscular dystrophies, myofibrillar myopathies, congenital myopathies, and Pompe disease.^{16,17} The presence of facial muscle weakness and the striking asymmetry of scapuloperoneal muscle weakness in this patient are highly suggestive of FSHD.

FACIOSCAPULOHUMERAL MUSCULAR DYSTROPHY

FSHD is the second most common muscular dystrophy in adults, following myotonic dystrophy type 1. It is caused by dominantly inherited or de novo mutations, and the estimated prevalence is 1 case per 15,000 to 20,000 persons. There are two types of FSHD, which are phenotypically indistinguishable. FSHD type 1 accounts for 95% of cases of FSHD and is caused by a contraction of the subtelomeric D4Z4 macrosatellite repeats on chromosome 4q35 in the presence of a permissive A allele. FSHD type 2 accounts for the other 5% of cases and is caused

by a mutation in *SMCHD1* (the gene that encodes the structural maintenance of chromosomes flexible hinge domain-containing protein 1) on chromosome 18p11.32 in the presence of a permissive A allele. Both genetic defects lead to chromatin relaxation (hypomethylation) of the permissive A allele in the D4Z4 region and result in expression of the double homeobox 4 (DUX4) protein, which is thought to be toxic (Fig. 2).^{18–23} DUX4 protein expression is a molecular target for FSHD therapies that are currently being investigated.

In patients with FSHD, the progression of muscle weakness usually follows a descending pattern, with weakness starting in facial muscles and then progressing to sequentially involve periscapular muscles, humeral muscles, abdominal and hip-girdle muscles, and finally distal muscles of the anterior compartment of the leg. Facial muscle weakness is often unnoticed, and the most common presenting symptom is difficulty with lifting objects above the shoulder, which usually occurs in the second or third decade of life. However, in some patients who do not have jobs or hobbies that are physically demanding, weakness of the shoulder-girdle muscles is unnoticed, even for decades, because of the absence of symptoms or compensation for symptoms. Such patients present later in life, when weakness of the distal leg muscles develops, as was seen in this case.^{23–25}

Although muscle weakness usually progresses slowly in patients with FSHD, periods of stability can be interrupted by periods of rapid deterioration of specific muscle groups. The trigger of these unexpected phases of decline is uncertain.^{24–26} I suspect that this patient's 5-year history of relatively rapid progression of left foot drop corresponds to a period of rapid deterioration of the muscles of the anterior compartment of the left leg, which interrupted the stability of his lifelong asymptomatic muscle weakness.

To determine whether this patient had longstanding muscle weakness that may have been unnoticed when he was younger, I would ask him whether he has ever kept his eyes partially open during sleep or had difficulty with whistling or drinking through a straw, lifting objects over his shoulders (e.g., reaching top cabinets), or performing abdominal crunches. In addition, I would look for horizontal axillary folds and Beevor's sign (upward deviation of the umbilicus on trun-

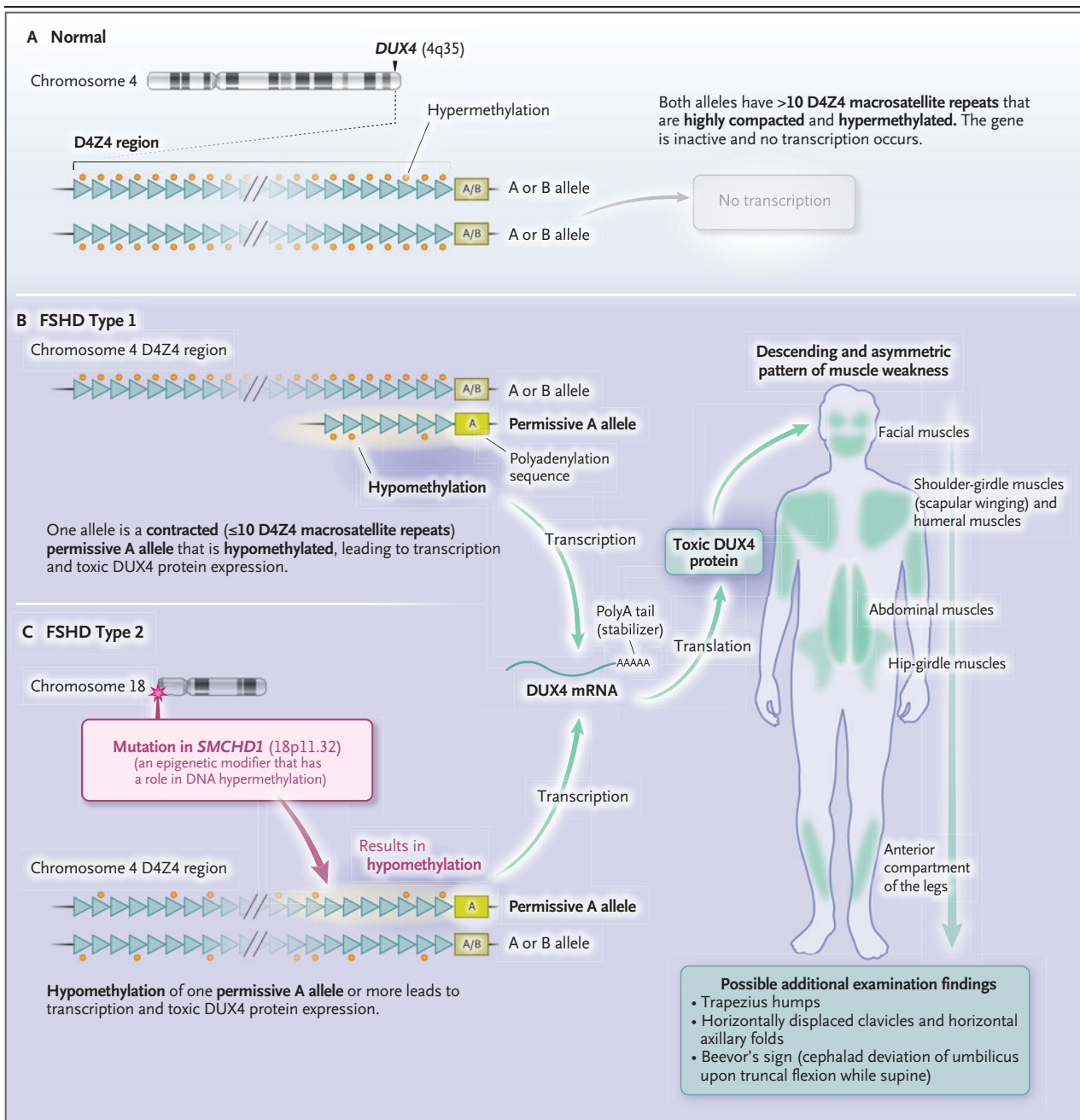


Figure 2. Genetic Basis and Clinical Characteristics of FSHD.

Genetic processes in a normal state (Panel A), in patients with facioscapulohumeral muscular dystrophy (FSHD) type 1 (Panel B), and in patients with FSHD type 2 (Panel C) are shown, along with clinical characteristics of FSHD. *DUX4* is the gene that encodes double homeobox 4 (DUX4); *SMCHD1* is the gene that encodes the structural maintenance of chromosomes flexible hinge domain-containing protein 1.

cal flexion while the patient is in a supine position),²⁷ which are manifestations of weakness in the pectoralis and abdominal muscles, respectively; these findings are not specific for FSHD but are typically seen in patients with this condition.

The next steps would be to provide genetic counseling and obtain consent for genetic testing for FSHD type 1. If testing for FSHD type 1 is negative, I would then test for FSHD type 2. In parallel, I would obtain repeat EMG studies

to look for evidence of a superimposed L5 radiculopathy.

The decrease in vibratory and pinprick sensations in the distal legs would not be explained by FSHD. I suspect that these findings correspond to a subclinical length-dependent sensory polyneuropathy that may be related to his alcohol use and is unrelated to his muscle weakness.

DR. PALOMA GONZÁLEZ-PÉREZ'S
DIAGNOSIS

Facioscapulohumeral muscular dystrophy.

DIAGNOSTIC TESTING

Dr. Reza Sadjadi: Electrodiagnostic studies, including nerve-conduction studies and EMG, were performed. The results of nerve-conduction studies recorded from the tibialis anterior muscle were notable for asymmetrically low amplitude of the motor response in the left common peroneal nerve. The differential diagnosis for this pattern, in conjunction with a relatively preserved sensory response in the superficial peroneal nerve, includes preganglionic neurogenic processes (typically L5 radiculopathy), neuromuscular-junction disorders (myasthenia), and myopathic processes. In addition, in patients with acute peripheral nerve injury (i.e., peroneal neuropathy), Wallerian degeneration of the motor fibers precedes that of the sensory fibers by a few days. Therefore, trauma to the peroneal nerve that occurred within 10 days before the nerve-conduction studies could explain normal sensory responses with low-amplitude motor responses.^{28,29}

EMG can help in differentiating between a primarily neurogenic process and a myopathic process (Fig. 3).³² In patients with a neurogenic process, reduced recruitment of motor-unit action potentials (with long duration and high amplitude) reflects the increased number of muscle fibers per functioning motor unit.³³ By contrast, in patients with a myopathic process, early recruitment of motor-unit action potentials (with short duration and low amplitude) reflects the decreased size and number of functional muscle fibers per motor unit.³⁴ EMG results recorded from the tibialis anterior muscle in this patient were supportive of a myopathic process.

Given the clinical phenotype and electrodiagnostic findings, targeted genetic testing was

performed. There was a contraction of the D4Z4 macrosatellite repeats on chromosome 4q35 leading to DUX4 protein expression, a finding consistent with FSHD type 1.³⁵ The patient confirmed long-standing difficulty with raising his right arm above his shoulder, whistling, and sipping through straws.

GENETIC DIAGNOSIS

Facioscapulohumeral muscular dystrophy.

DISCUSSION OF MANAGEMENT

Dr. Sadjadi: Treatment for FSHD is centered on the management of muscle weakness and extramuscular manifestations of the disease. Aerobic exercises have been shown to be beneficial in patients with FSHD, although not in patients with other myopathic processes.³⁶ After diagnosis, this patient began attending exercise classes for core strengthening and balance training. The use of a ground-reaction ankle-foot orthosis has been shown to be particularly helpful for the reduction of knee buckling in patients with concomitant weakness on knee flexion.²³ In some patients with relatively preserved deltoid strength, scapular fixation of the dominant side is performed for the temporary improvement of range of motion^{37,38}; this patient was not a candidate for this treatment.

Extramuscular manifestations of FSHD that are associated with large D4Z4 repeat deletions include hearing loss and exudative retinopathy (Coats' disease).²³ This patient underwent audiometry screening^{35,36,39}; he did not have any hearing loss. He was referred to the ophthalmology clinic and was advised to use eye lubricants. Eye taping at night and surgical insertion of lid weights can be performed for the prevention of xerophthalmia and corneal abrasion.⁴⁰ Fluorescein angiography can be used for the identification of retinal vasculopathy, and laser photocoagulation may stop progression to full Coats' disease and vision loss.^{41,42} In addition, if FSHD leads to restrictive lung disease, the use of bi-level positive airway pressure at night may be considered.^{43,44}

A follow-up examination performed in this patient 10 years after the initial assessment was notable for moderate weakness in the eyelids with normal extraocular movements. He had asymmetric shoulder shrug, with less rise on the

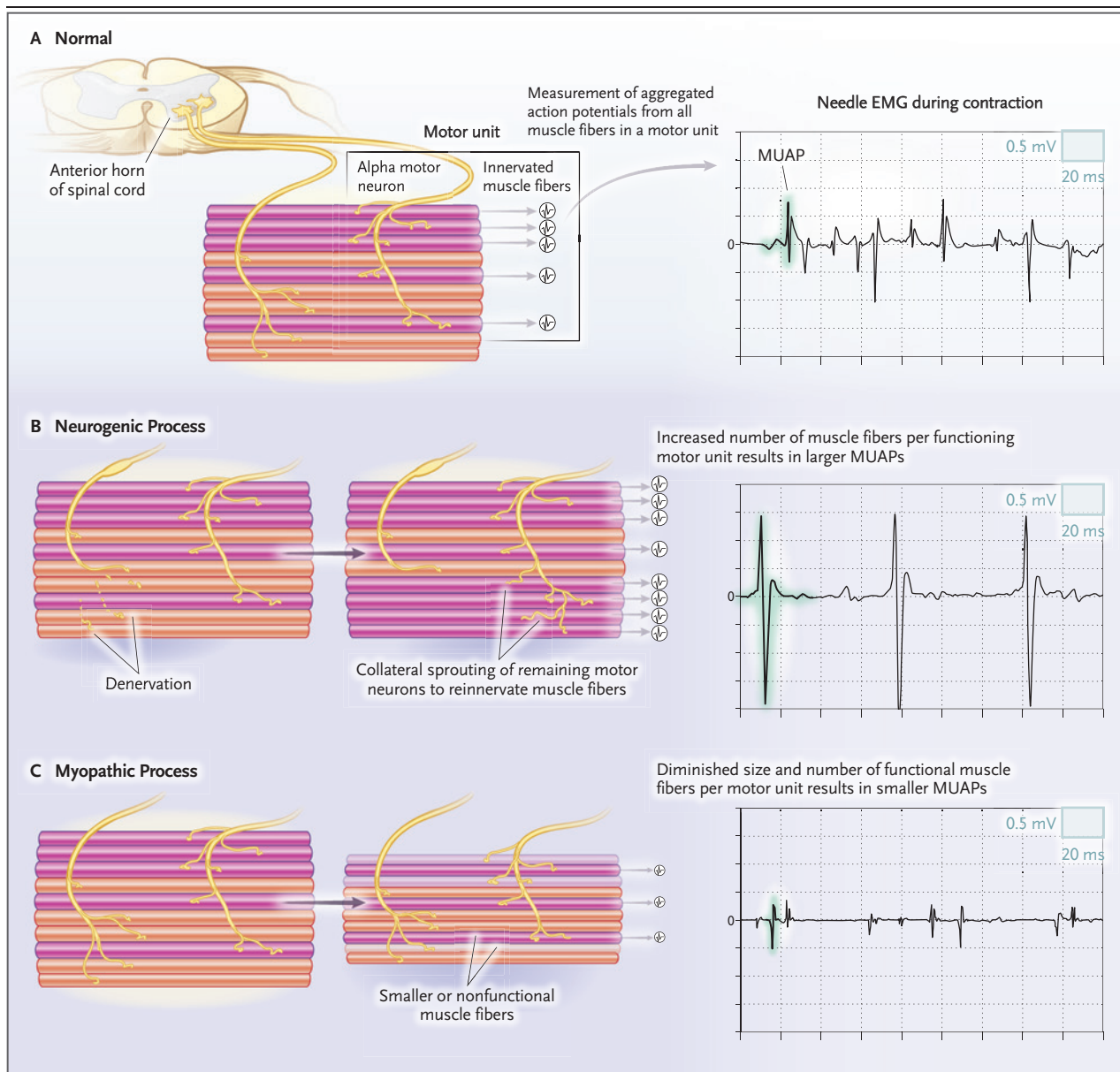


Figure 3. Normal, Neurogenic, and Myopathic Patterns on Needle EMG.

Characteristics of normal (Panel A), neurogenic (Panel B), and myopathic (Panel C) motor-unit action potentials (MUAPs) observed on needle electromyography (EMG) are shown.³⁰⁻³⁴ Reduced recruitment of reinnervated MUAPs (with long duration and high amplitude) is characteristic of a neurogenic process (Panel B). Early recruitment of brief MUAPs (with short duration and low amplitude) is characteristic of a myopathic process (Panel C). Early recruitment refers to the generation of excess MUAPs as compensation for diminished functional motor units.

left side. Wasting of the shoulder-girdle muscles was more prominent on the right side, where his scapular winging was more severe. He had mild weakness on hip flexion and knee flexion on both sides, on dorsiflexion of the right ankle, and on antigravity dorsiflexion and plantar flexion of the left ankle.

FINAL DIAGNOSIS

Facioscapulohumeral muscular dystrophy.

This case was presented at Neurology Grand Rounds. Disclosure forms provided by the authors are available with the full text of this article at NEJM.org.

We thank Dr. William S. David for his assistance with preparation of the case presentation.

REFERENCES

- Ozdemir N, Citak G, Acar UD. Spastic foot drop caused by a brain tumour: a case report. *Br J Neurosurg* 2004;18:314-5.
- Westhout FD, Paré LS, Linskey ME. Central causes of foot drop: rare and underappreciated differential diagnoses. *J Spinal Cord Med* 2007;30:62-6.
- Alhardallo M, El Ansari W, Baco AM. Second ever reported case of central cause of unilateral foot drop due to cervical disc herniation: case report and review of literature. *Int J Surg Case Rep* 2021;83:105928.
- Takagi H, Kawaguchi Y, Kanamori M, Abe Y, Kimura T. T1-2 disc herniation following an en bloc cervical laminoplasty. *J Orthop Sci* 2002;7:495-7.
- Tokuhashi Y, Matsuzaki H, Uematsu Y, Oda H. Symptoms of thoracolumbar junction disc herniation. *Spine (Phila Pa 1976)* 2001;26(22):E512-E518.
- Zhang C, Xue Y, Wang P, Yang Z, Dai Q, Zhou H-F. Foot drop caused by single-level disc protrusion between T10 and L1. *Spine (Phila Pa 1976)* 2013;38:2295-301.
- Aquino CC, Slow E, Lang AE. Dystonic pseudo foot drop. *Mov Disord Clin Pract* 2015;2:295-8.
- de Schipper LJ, Boon AJW, Muntz AG. Foot drop dystonia resulting from *parkin* (PARK2) mutation. *Mov Disord Clin Pract* 2015;2:292-4.
- Milanov I, Ishpekova B. Differential diagnosis of scapulo-peroneal syndrome. *Electromyogr Clin Neurophysiol* 1997;37:73-8.
- Ricker K, Mertens HG, Schimrigk K. The neurogenic scapulo-peroneal syndrome. *Eur Neurol* 1968;1:257-74.
- Biasini F, Portaro S, Mazzeo A, et al. TRPV4 related scapulo-peroneal spinal muscular atrophy: report of an Italian family and review of the literature. *Neuromuscul Disord* 2016;26:312-5.
- Koutsis G, Lynch D, Manole A, et al. Charcot-Marie-Tooth disease type 2C and scapulo-peroneal muscular atrophy overlap syndrome in a patient with the R232C TRPV4 mutation. *J Neurol* 2015;262:1972-5.
- Krurup C. Pitfalls in electrodiagnosis. *J Neurol* 1999;246:1115-26.
- Liguori R, Fuglsang-Frederiksen A, Nix W, Fawcett PR, Andersen K. Electromyography in myopathy. *Neurophysiol Clin* 1997;27:200-3.
- Nojszewska M, Gawel M, Kierdaszuk B, et al. Electromyographic findings in sporadic inclusion body myositis. *J Electromyogr Kinesiol* 2018;39:114-9.
- Barohn RJ, Dimachkie MM, Jackson CE. A pattern recognition approach to patients with a suspected myopathy. *Neurol Clin* 2014;32:569-93.
- Barohn RJ, McVey AL, DiMauro S. Adult acid maltase deficiency. *Muscle Nerve* 1993;16:672-6.
- Wijmenga C, Padberg GW, Moerer P, et al. Mapping of facioscapulohumeral muscular dystrophy gene to chromosome 4q35-qter by multipoint linkage analysis and in situ hybridization. *Genomics* 1991;9:570-5.
- van Deutekom JC, Wijmenga C, van Tienhoven EA, et al. FSHD associated DNA rearrangements are due to deletions of integral copies of a 3.2 kb tandemly repeated unit. *Hum Mol Genet* 1993;2:2037-42.
- van Overveld PGM, Lemmers RJFL, Sandkuijl LA, et al. Hypomethylation of D4Z4 in 4q-linked and non-4q-linked facioscapulohumeral muscular dystrophy. *Nat Genet* 2003;35:315-7.
- Lemmers RJFL, de Kievit P, Sandkuijl L, et al. Facioscapulohumeral muscular dystrophy is uniquely associated with one of the two variants of the 4q subtelomere. *Nat Genet* 2002;32:235-6.
- Lemmers RJFL, O'Shea S, Padberg GW, Lunt PW, van der Maarel SM. Best practice guidelines on genetic diagnostics of Facioscapulohumeral muscular dystrophy: workshop 9th June 2010, LUMC, Leiden, the Netherlands. *Neuromuscul Disord* 2012;22:463-70.
- Tawil R. Facioscapulohumeral muscular dystrophy. *Handb Clin Neurol* 2018;148:541-8.
- Statland JM, McDermott MP, Heatwole C, et al. Reevaluating measures of disease progression in facioscapulohumeral muscular dystrophy. *Neuromuscul Disord* 2013;23:306-12.
- Richards M, Coppée F, Thomas N, Belayew A, Upadhyaya M. Facioscapulohumeral muscular dystrophy (FSHD): an enigma unravelled? *Hum Genet* 2012;131:325-40.
- Amato AA, Russell JA. *Neuromuscular disorders*, 2nd ed. New York: McGraw-Hill Education, 2016.
- Althagafi A, Nadi M. Beever Sign. In: StatPearls. Treasure Island, FL: StatPearls Publishing, 2022 (<http://www.ncbi.nlm.nih.gov/books/NBK545262/>).
- Gutmann E, Holubar J. The degeneration of peripheral nerve fibers. *J Neurol Neurosurg Psychiatry* 1950;13:89-105.
- Chaudhry V, Glass JD, Griffin JW. Wallerian degeneration in peripheral nerve disease. *Neurol Clin* 1992;10:613-27.
- Eccles JC, Sherrington CS. Numbers and contraction-values of individual motor-units examined in some muscles of the limb. *Proc R Soc Lond Ser B Contain Pap Biol Character* 1930;106:326-57.
- Wohlfart G. Collateral regeneration from residual motor nerve fibers in amyotrophic lateral sclerosis. *Neurology* 1957;7:124-34.
- Preston D, Shapiro B. *Electromyography and neuromuscular disorders*, 4th ed. Amsterdam: Elsevier, 2020 (<https://shop.elsevier.com/books/electromyography-and-neuromuscular-disorders/preston/978-0-323-66180-5>).
- Mills KR. The basics of electromyography. *J Neurol Neurosurg Psychiatry* 2005;76:Suppl 2:ii32-ii35.
- Bromberg MB. The motor unit and quantitative electromyography. *Muscle Nerve* 2020;61:131-42.
- Preston MK, Tawil R, Wang LH. Facioscapulohumeral muscular dystrophy. In: Adam MP, Everman DB, Mirza GM, et al., eds. *GeneReviews*. Seattle: University of Washington, 1993 (<http://www.ncbi.nlm.nih.gov/books/NBK1443/>).
- Voet NB, van der Kooi EL, van Engelen BG, Geurts AC. Strength training and aerobic exercise training for muscle disease. *Cochrane Database Syst Rev* 2019;12(12):CD003907.
- Giannini S, Ceccarelli F, Faldini C, Pagkrati S, Merlini L. Scapulopexy of winged scapula secondary to facioscapulohumeral muscular dystrophy. *Clin Orthop Relat Res* 2006;449:288-94.
- Orrell RW, Copeland S, Rose MR. Scapular fixation in muscular dystrophy. *Cochrane Database Syst Rev* 2010;2010(1):CD003278.
- Kamata T, Miyajima H, Shimizu T, Fujimoto M, Takahashi Y, Kaneko E. Facioscapulohumeral syndrome with sensorineural hearing loss and abnormality of retinal vessels. *Intern Med* 1993;32:678-80.
- Sansone V, Boynton J, Palenski C. Use of gold weights to correct lagophthalmos in neuromuscular disease. *Neurology* 1997;48:1500-3.
- Desai UR, Sabates FN. Long-term follow-up of facioscapulohumeral muscular dystrophy and Coats' disease. *Am J Ophthalmol* 1990;110:568-9.
- Pauleikhoff D, Bornfeld N, Bird AC, Wessing A. Severe visual loss associated with retinal telangiectasis and facioscapulohumeral muscular dystrophy. *Graefes Arch Clin Exp Ophthalmol* 1992;230:362-5.
- Santos DR, Boussaid G, Stojkovic T, et al. Respiratory muscle dysfunction in facioscapulohumeral muscular dystrophy: reference article: sleep-related breathing disorders in facioscapulohumeral dystrophy. *Sleep Breath* 2020;24:603-4.
- Runte M, Spiesshoefer J, Heidbreder A, et al. Sleep-related breathing disorders in facioscapulohumeral dystrophy. *Sleep Breath* 2019;23:899-906.

Copyright © 2023 Massachusetts Medical Society.